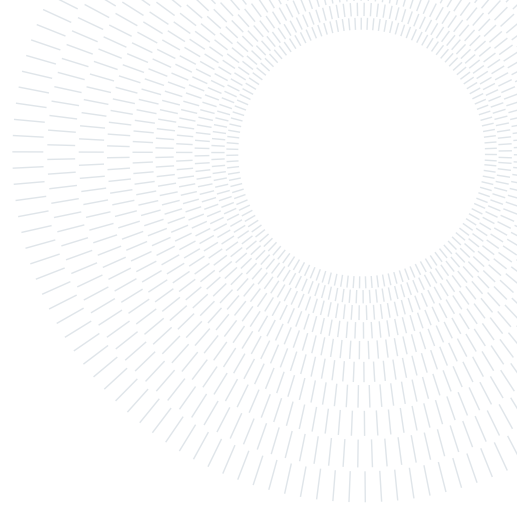




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# Pneumonia Detection Using Anchor-Free Object Detection

REPORT

NUMERICAL ANALYSIS FOR MACHINE LEARNING

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**Abstract:** This project aims to re-implement and investigate an anchor-free object detection framework for pneumonia localization in chest X-ray images. The work is motivated by the framework proposed for pneumonia detection on the RSNA dataset, while adopting FCOS as the main reference for the design of the detector. The resulting architecture combines a ResNet backbone with a Feature Pyramid Network (FPN) and detection heads for classification, bounding-box regression, and centerness prediction. The implementation is evaluated using both ImageNet- and Chest-Xray-pretrained backbones and considers different ResNet depths. Particular attention is given to training stability, leading to the adoption of training strategies including learning-rate scheduling, backbone freezing and gradual unfreezing, gradient clipping, and exponential moving average (EMA) of the model parameters. The final system therefore preserves the medical application and dataset setting of the reference pneumonia study while following the anchor-free, fully convolutional formulation of FCOS, with additional implementation choices introduced to improve robustness and flexibility in the target application.

**Key-words:** anchor-free object detection, FCOS, pneumonia detection, chest X-ray, ResNet, transfer learning

## 1. Introduction

Pneumonia is a major respiratory disease and remains an important cause of morbidity and mortality worldwide. Chest X-ray imaging is one of the most widely used tools for its assessment, but the interpretation of radiographs is challenging because the manifestations of pneumonia can be subtle, heterogeneous, and spatially overlapping with other abnormalities. In addition to determining whether pneumonia is present, identifying its precise location is particularly relevant for computer-assisted analysis, as the affected regions may vary substantially in size, shape, and appearance. The availability of the RSNA Pneumonia Detection Challenge dataset has made it possible to investigate deep learning methods for the automated localization of pneumonia in chest radiographs. In particular, Wu et al. [1] proposed an anchor-free object detection framework for this task, demonstrating the potential of fully convolutional detection methods in the medical imaging domain.

Object detection traditionally relies on predefined anchor boxes, whose dimensions, aspect ratios, and scales must be selected before training. These design choices introduce additional hyperparameters and can be particularly restrictive when the target objects exhibit large variations in size and shape. Anchor-free detectors address this issue by removing the need for predefined boxes and directly predicting object information from spatial locations in the feature maps. Among these approaches, Fully Convolutional One-Stage Object Detection (FCOS) formulates detection as a dense, per-location prediction problem, in which each spatial location is associated with a bounding-box regression and a classification prediction, together with a centerness score used

to suppress low-quality detections [2]. This formulation is especially interesting in the context of pneumonia localization, where the shape and extent of the detected regions may vary considerably.

The objective of this project is to re-implement the pneumonia detection framework proposed by Wu et al. [1], with a particular focus on the FCOS detection paradigm. Rather than reproducing the reference implementation in every detail, the project develops an FCOS-based detector and investigates the effect of different backbone configurations. The detector is built upon a ResNet backbone [3] and a Feature Pyramid Network (FPN) [4], which provides feature representations at multiple spatial scales. The resulting feature maps are processed by FCOS-style detection heads that predict classification, bounding-box regression, and centerness at each pyramid level. Focal Loss [5] is employed to address the severe foreground-background imbalance arising from dense prediction.

A further objective of the project is to investigate how the initialization and depth of the backbone affect the resulting detector. Four backbone configurations are considered: ResNet-50 and ResNet-101 initialized with ImageNet-pretrained weights, and ResNet-50 and ResNet-101 initialized through a preliminary Chest-Xray classification pretraining stage. In the latter case, the ResNet is first trained to distinguish normal from pneumonia chest radiographs and is subsequently reused as the feature extractor of the detection network. This allows the study to distinguish the effect of increasing network depth from that of adapting the learned representation to the target medical domain.

The implementation also investigates training strategies aimed at improving optimization stability. In particular, the training procedure includes learning-rate scheduling, temporary backbone freezing followed by gradual unfreezing, gradient clipping, and an exponential moving average of the trainable parameters. These choices are motivated by the optimization difficulties observed during the development of the detector and are intended to provide a more stable training procedure while preserving the FCOS formulation as the main detection reference.

The main objectives of this project can therefore be summarized as follows:

- re-implement an anchor-free pneumonia detection system based on the RSNA dataset and the FCOS formulation;
- investigate the role of the backbone architecture by comparing ResNet-50 and ResNet-101;
- compare generic ImageNet pretraining with domain-specific Chest-Xray pretraining;
- analyze the effects of different training strategies adopted to improve optimization stability.

## 2. Dataset

Two different datasets are used in this project for two distinct stages of the pipeline. The Chest-Xray dataset is employed during the preliminary classification pretraining of the ResNet backbone, whereas the RSNA Pneumonia Detection dataset is used for training and evaluating the final object detector.

### 2.1. Chest-Xray Classification Dataset

The first dataset is used to pretrain the ResNet backbone on a binary classification task. The dataset is organized into three independent splits: `train`, `val`, and `test`. Each split contains two class directories, `NORMAL` and `PNEUMONIA`, as illustrated by the following structure:

```
data/chest_xray/
|-- train/
|   |-- NORMAL/
|   '-- PNEUMONIA/
|-- val/
|   |-- NORMAL/
|   '-- PNEUMONIA/
'-- test/
    |-- NORMAL/
    '-- PNEUMONIA/
```

The dataset contains 5,856 chest X-ray images in total. The training, validation, and test sets contain 4,684, 585, and 587 images, respectively. The class distribution is highly consistent across the three splits, with approximately 73% of the images belonging to the `PNEUMONIA` class and 27% to the `NORMAL` class.

Table 1: Chest-Xray class distribution.

| Split        | NORMAL      | PNEUMONIA   | Total       | PNEUMONIA (%) |
|--------------|-------------|-------------|-------------|---------------|
| Train        | 1266        | 3418        | 4684        | 72.97         |
| Validation   | 158         | 427         | 585         | 72.99         |
| Test         | 159         | 428         | 587         | 72.91         |
| <b>Total</b> | <b>1583</b> | <b>4273</b> | <b>5856</b> | <b>72.97</b>  |

The dataset is therefore imbalanced towards the PNEUMONIA class, with approximately 2.7 pneumonia images for every normal image.

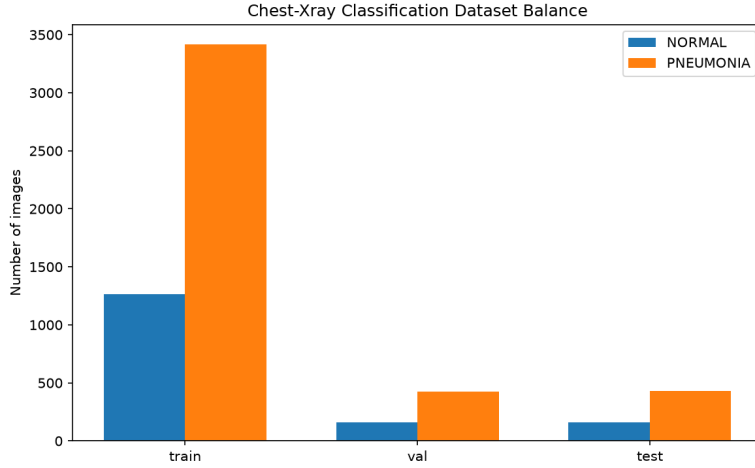


Figure 1: Chest-Xray class distribution across the three splits.

## 2.2. RSNA Pneumonia Detection Dataset

The second dataset is the RSNA Pneumonia Detection Challenge dataset, which is used for the object detection task. It contains chest radiographs stored as DICOM files together with a separate CSV file containing the corresponding pneumonia annotations.

The dataset is organized as follows:

```
data/rsna-pneumonia-detection-challenge/
|-- stage_2_train_images/
|   |-- <patientId>.dcm
|   |-- <patientId>.dcm
|   '-- ...
'-- stage_2_train_labels.csv
```

A total of 26,684 DICOM images are present in the dataset. According to the annotation file, 6,012 images contain at least one positive pneumonia annotation, while 20,672 images do not contain a positive pneumonia annotation. This corresponds to approximately 22.53% positive images and 77.47% negative images.

Table 2: RSNA image-level class distribution.

| Category     | Number of images | Percentage     |
|--------------|------------------|----------------|
| Positive     | 6012             | 22.53%         |
| Negative     | 20672            | 77.47%         |
| <b>Total</b> | <b>26684</b>     | <b>100.00%</b> |

The RSNA dataset is therefore strongly imbalanced at the image level, with most images containing no positive pneumonia annotation. This distribution should be distinguished from the foreground-background imbalance introduced by dense object detection, which is discussed separately in Section 5.

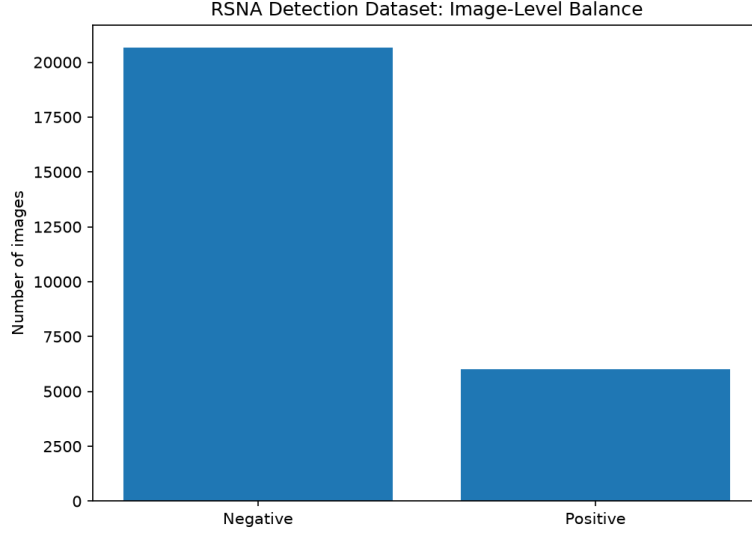


Figure 2: Positive and negative RSNA images.

### 3. Network Architecture

The proposed detector is composed of three main components: a ResNet backbone, a Feature Pyramid Network (FPN), and a set of detection heads. The overall architecture follows the fully convolutional and anchor-free formulation of FCOS [2], while the individual components are implemented and adapted to the pneumonia detection task.

#### 3.1. ResNet Backbone

The first component of the network is a Residual Network (ResNet), introduced by He et al. [3]. The main motivation behind ResNet is the difficulty of optimizing very deep neural networks. In particular, the original work identified a degradation problem in which increasing the depth of a conventional “plain” network could lead to higher training error rather than improved performance. The authors showed that this behaviour was not simply caused by overfitting, and that the problem could persist even in the presence of Batch Normalization and apparently well-behaved gradient norms. They therefore interpreted the phenomenon primarily as an optimization difficulty associated with very deep networks.

The key idea introduced by ResNet is residual learning. Instead of directly learning a desired mapping  $H(x)$ , a stack of layers is trained to learn the residual function

$$F(x) = H(x) - x, \quad (1)$$

so that the original mapping can be written as

$$H(x) = F(x) + x. \quad (2)$$

This formulation is implemented through a shortcut connection that directly forwards the input to the output of the residual branch. According to the authors, this reformulation makes the optimization easier, particularly when the desired transformation is close to an identity mapping. In such cases, the residual branch can approach zero rather than requiring a stack of non-linear layers to explicitly learn an identity transformation [3].

##### 3.1.1. Bottleneck Residual Block

The ResNet-50 and ResNet-101 models used in this project employ the bottleneck residual block introduced in the original ResNet architecture. Unlike the basic two-layer residual block, the bottleneck block consists of three convolutions:

1. a  $1 \times 1$  convolution that projects the input into a lower dimensional feature space;
2. a  $3 \times 3$  convolution that performs the main spatial feature extraction;
3. a final  $1 \times 1$  convolution that restores the expanded channel dimension.

For an input with  $C$  channels, the bottleneck representation uses an intermediate dimensionality of  $C/4$  in the conceptual bottleneck view, while the final convolution expands the representation by a factor of four. In the implementation used in this project, this expansion is explicitly defined as

$$C_{\text{out}} = 4C_{\text{bottleneck}}. \quad (3)$$

The three convolutions are combined with Batch Normalization and ReLU activations, while the original input is propagated through a shortcut branch. The residual output is therefore computed as

$$y = \text{ReLU}(F(x) + S(x)), \quad (4)$$

where  $F(x)$  denotes the three-layer residual branch and  $S(x)$  denotes the shortcut. When the input and output dimensions are compatible,  $S(x)$  is an identity mapping. When their dimensions differ, a projection shortcut based on a  $1 \times 1$  convolution and Batch Normalization is used to match the dimensions [3].

### 3.1.2. ResNet-50 and ResNet-101 Structure

The main difference between ResNet-50 and ResNet-101 is the number of bottleneck blocks stacked at each stage. Following the architecture reported by He et al. [3], the four residual stages are organized as

$$\text{ResNet-50} = [3, 4, 6, 3] \quad (5)$$

and

$$\text{ResNet-101} = [3, 4, 23, 3]. \quad (6)$$

| layer name         | output size | 18-layer                                                                    | 34-layer                                                                    | 50-layer                                                                                        | 101-layer                                                                                        | 152-layer                                                                                        |
|--------------------|-------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| conv1              | 112×112     | 7×7, 64, stride 2                                                           |                                                                             |                                                                                                 |                                                                                                  |                                                                                                  |
| conv2 <sub>x</sub> | 56×56       | 3×3 max pool, stride 2                                                      |                                                                             |                                                                                                 |                                                                                                  |                                                                                                  |
|                    |             | $\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$   | $\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 3$   | $\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$    | $\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$     | $\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$     |
| conv3 <sub>x</sub> | 28×28       | $\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$ | $\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 4$ | $\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$  | $\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$   | $\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 8$   |
| conv4 <sub>x</sub> | 14×14       | $\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$ | $\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 6$ | $\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$ | $\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 23$ | $\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 36$ |
| conv5 <sub>x</sub> | 7×7         | $\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$ | $\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 3$ | $\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$ | $\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$  | $\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$  |
|                    | 1×1         | average pool, 1000-d fc, softmax                                            |                                                                             |                                                                                                 |                                                                                                  |                                                                                                  |
| FLOPs              |             | $1.8 \times 10^9$                                                           | $3.6 \times 10^9$                                                           | $3.8 \times 10^9$                                                                               | $7.6 \times 10^9$                                                                                | $11.3 \times 10^9$                                                                               |

Figure 3: ResNet-50 and ResNet-101 stage structure.

The number of blocks is therefore increased almost entirely in the third residual stage. The resulting feature representations have the following channel dimensions:

| Stage              | Feature | Channels | ResNet-50 / 101 blocks |
|--------------------|---------|----------|------------------------|
| Initial            | —       | 64       | —                      |
| conv2 <sub>x</sub> | $C_2$   | 256      | 3 / 3                  |
| conv3 <sub>x</sub> | $C_3$   | 512      | 4 / 4                  |
| conv4 <sub>x</sub> | $C_4$   | 1024     | 6 / 23                 |
| conv5 <sub>x</sub> | $C_5$   | 2048     | 3 / 3                  |

The spatial resolution is progressively reduced throughout the network, while the number of feature channels increases. This hierarchical structure is particularly suitable for detection because it provides feature representations at different levels of abstraction.

The original ResNet paper reports the same bottleneck configurations for ResNet-50 and ResNet-101, with the corresponding difference in depth and computational cost [3].

### 3.1.3. Implementation

Two implementations of the ResNet backbone were used during the development and evaluation of the detection pipeline. The first relies on the standard implementations provided by `torchvision`, which were used to obtain the established ResNet-50 and ResNet-101 architectures and, in particular, their ImageNet-pretrained weights. In parallel, a custom PyTorch implementation of ResNet was developed to provide direct control over the residual blocks, intermediate stage outputs, and parameter loading. This custom implementation was useful for inspecting and validating the internal operations of the backbone, testing different configurations, and integrating the selected feature maps directly with the Feature Pyramid Network (FPN).

The custom implementation follows the standard ResNet architecture, and the same implementation is used for both depths; only the number of bottleneck blocks differs between ResNet-50 and ResNet-101. The use of both the `torchvision` implementation and the custom implementation also provided a way to cross-check the backbone behaviour while retaining sufficient flexibility for subsequent experiments and modifications to the detection pipeline.

## 3.2. Feature Pyramid Network

The second component of the detector is the Feature Pyramid Network (FPN), introduced by Lin et al. [4]. The main motivation behind FPN is the need to represent objects at different spatial scales while retaining strong semantic information at every resolution.

A convolutional backbone naturally produces a hierarchical representation of the input image. As the network progresses towards deeper layers, repeated downsampling reduces the spatial resolution of the feature maps, while the features become increasingly rich in high-level semantic information. Conversely, shallower layers preserve more accurate spatial and localization information, but their representations are semantically weaker. The original FPN paper refers to this discrepancy as a semantic gap between the different levels of the feature hierarchy [4].

This creates an inherent trade-off for object detection. Using only deep features provides strong semantic representations but poor spatial resolution, which can be particularly problematic for small objects. On the other hand, using only shallow features preserves spatial detail but provides less discriminative semantic information. The purpose of FPN is therefore not simply to obtain feature maps at different scales, but to enrich the high-resolution maps with the semantic information available in deeper layers [4].

### 3.2.1. FPN Construction

For a ResNet backbone, the original FPN identifies the outputs of the last residual block of each backbone stage as  $C_2$ ,  $C_3$ ,  $C_4$ , and  $C_5$ . These feature maps have strides of 4, 8, 16, and 32 pixels with respect to the input image, respectively [4].

The FPN combines these feature maps through two complementary mechanisms. The first is a top-down pathway, which propagates information from deeper, coarser feature maps towards shallower, finer ones. The second consists of lateral connections, which inject the spatially more precise information of the corresponding backbone features into the top-down pathway.

The construction starts from the deepest feature map. A  $1 \times 1$  convolution first projects  $C_5$  into the feature dimension used by the pyramid. The resulting representation is then progressively upsampled by a factor of two and added to a lateral projection of the feature map at the next level. A  $3 \times 3$  convolution is finally applied to the merged feature map in order to produce the output pyramid level.

Conceptually, the process can be expressed as

$$P_5 = \text{Conv}_{1 \times 1}(C_5), \quad (7)$$

followed by

$$P_4 = \text{Conv}_{3 \times 3}(\text{Conv}_{1 \times 1}(C_4) + \text{Upsample}(P_5)), \quad (8)$$

and

$$P_3 = \text{Conv}_{3 \times 3}(\text{Conv}_{1 \times 1}(C_3) + \text{Upsample}(P_4)). \quad (9)$$

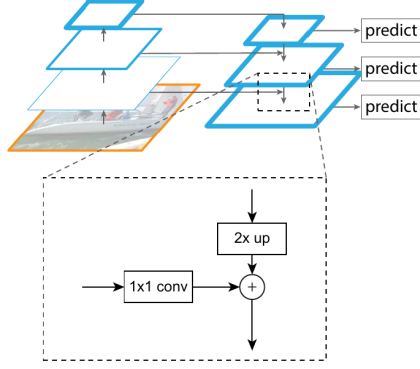


Figure 4: Feature Pyramid Network.

The lateral terms preserve the spatial information of the shallower features, while the top-down pathway transfers the stronger semantic representation of the deeper features to higher resolutions. The resulting feature maps therefore contain a more balanced combination of semantic richness and spatial precision. The original FPN paper fixes the number of channels of the pyramid features to 256 and uses the same feature dimension at all levels [4].

### 3.2.2. FPN in the Implemented Detector

The implementation developed in this project follows the same top-down and lateral-connection principle, while adapting the pyramid to the requirements of FCOS. The ResNet backbone produces four outputs,

$$\{C_2, C_3, C_4, C_5\}, \quad (10)$$

but  $C_2$  is not used in the construction of the detection pyramid. The remaining feature maps  $C_3$ ,  $C_4$ , and  $C_5$  are projected to 256 channels through  $1 \times 1$  convolutions and combined through the top-down pathway.

In the implementation, the construction therefore proceeds as follows:

$$C_5 \rightarrow P_5, \quad (11)$$

$$C_4 + \text{Upsample}(P_5) \rightarrow P_4, \quad (12)$$

$$C_3 + \text{Upsample}(P_4) \rightarrow P_3. \quad (13)$$

The resulting three feature maps are then extended by two additional levels.  $P_6$  is generated using a  $3 \times 3$  convolution with stride 2 applied to  $P_5$ , while  $P_7$  is obtained by applying another stride-2 convolution to  $P_6$  after a ReLU activation.

Consequently, the final feature pyramid used by the detector is

$$\{P_3, P_4, P_5, P_6, P_7\}, \quad (14)$$

with corresponding strides

$$\{8, 16, 32, 64, 128\}. \quad (15)$$

This five-level configuration is particularly suitable for FCOS because different object sizes can be handled at different levels of the pyramid. The finer levels provide a denser spatial representation, whereas the coarser levels provide a larger receptive field and are therefore more appropriate for larger objects.

### 3.2.3. Relation to FPN and FCOS

The implemented pyramid follows FCOS rather than reproducing the feature construction of Wu et al. [1]. The reference framework uses five backbone outputs, projected independently to feature levels with strides 8, 16, 32, 64, 128, whereas the present detector builds  $P_3$ – $P_5$  through the top-down FPN pathway and derives  $P_6$  and  $P_7$  with stride-2 convolutions. Thus, the number and strides of the levels are shared, but the feature fusion is different. This is the main architectural choice that makes the implementation closer to FCOS than to a literal re-implementation of the reference pneumonia detector [2].



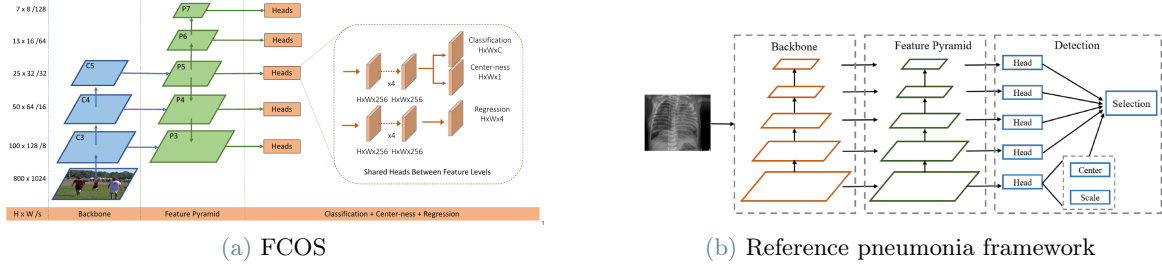


Figure 5: FCOS and reference pneumonia feature pyramids.

### 3.2.4. Architectural Choice

The choice to follow FCOS rather than reproduce the reference pneumonia architecture exactly is central to the project. The original FPN builds a top-down hierarchy by combining deeper semantic features with lateral connections from shallower layers [4]. FCOS uses this principle with  $P_3$ – $P_5$  obtained from  $C_3$ – $C_5$  and extends the pyramid with  $P_6$  and  $P_7$  through stride-2 convolutions [2]. This is the same feature-level structure used here.

The reference pneumonia framework also operates on five levels with strides 8, 16, 32, 64, 128, but constructs the levels from five backbone outputs that are projected independently [1]. Thus the two approaches share the multi-scale idea and the same nominal stride progression, but not the internal feature fusion. For this project, the FCOS construction was preferred because the goal was to study the anchor-free detector itself and not simply reproduce the reference implementation as a black box.

This distinction is important when interpreting the experiments: the results evaluate a specific FCOS-style detector applied to the RSNA pneumonia task, rather than a strict reproduction of every architectural and optimization choice made by Wu et al. [1].

### 3.3. Detection Head

The detection head converts the feature maps produced by the FPN into the quantities required by an anchor-free detector. Since no anchor boxes are used, the predictions are associated directly with spatial locations on each pyramid level. The implemented head follows the FCOS decomposition into classification, bounding-box regression and centerness, while keeping a simpler internal structure.

For each of the five levels  $P_3$ – $P_7$ , a separate **DetectionHead** module is instantiated. Each input feature map has 256 channels. The head contains two feature towers. The classification tower applies a  $3 \times 3$  convolution, Group Normalization and ReLU, followed by a  $1 \times 1$  convolution that produces one classification logit per location. The second tower has the same  $3 \times 3$ -GroupNorm-ReLU structure and is used jointly by the regression and centerness branches. Two  $1 \times 1$  convolutions then produce the four LTRB regression values and one centerness value.

For a pyramid level  $P_i$ , the classification features can be written as

$$F_{\text{cls}} = \text{ReLU}(\text{GN}(\text{Conv}_{3 \times 3}(P_i))), \quad (16)$$

with classification output

$$C_i = \text{Conv}_{1 \times 1}^{\text{cls}}(F_{\text{cls}}). \quad (17)$$

The regression tower is computed independently as

$$F_{\text{reg}} = \text{ReLU}(\text{GN}(\text{Conv}_{3 \times 3}(P_i))), \quad (18)$$

from which

$$R_i = \text{Conv}_{1 \times 1}^{\text{reg}}(F_{\text{reg}}), \quad S_i = \text{Conv}_{1 \times 1}^{\text{ctr}}(F_{\text{reg}}) \quad (19)$$

are obtained. Here  $C_i$ ,  $R_i$  and  $S_i$  denote classification, regression and centerness predictions, respectively.

This structure is close to FCOS at the functional level, since classification and localization are explicitly separated and centerness is predicted from the regression features [2]. There are, however, two relevant differences. First, the present implementation uses the same regression feature representation for both LTRB regression and centerness. Second, the head parameters are not shared across pyramid levels: five independent **DetectionHead**



modules are used. These are implementation choices rather than architectural claims about the original FCOS model.

The reference pneumonia detector also uses five feature levels, but its head is organized as a classification/localization module with a different scale selection mechanism [1]. The present implementation keeps the FCOS-style target assignment and centerness formulation instead. This makes the distinction between the project and a literal re-implementation of the reference model explicit.

The final output is therefore one classification value, four regression values and one centerness value at each spatial location of every FPN level. Although the current task is binary pneumonia detection, this separation would also allow the classification branch to be extended to multiple lesion classes without changing the localization representation.

## 4. Target Assignment

In an anchor-free detector, the network does not regress bounding boxes relative to predefined anchors. Instead, predictions are generated at the spatial locations of each FPN feature map. The target generation stage is therefore responsible for determining which locations correspond to ground-truth objects and for converting the associated bounding boxes into the representation expected by the detection heads.

For each FPN level, the target generator receives the ground-truth bounding boxes, the spatial dimensions of the corresponding feature map, and its stride. It produces three target tensors: a binary positive mask, the four LTRB regression distances, and a centerness target. These targets are then used by the corresponding training losses.

### 4.1. Target Generation

The target assignment procedure follows the main formulation of FCOS [2]. First, every spatial location of the feature map is mapped back to the coordinate system of the original image. For a feature map location  $(x, y)$  and a stride  $s$ , the corresponding image-space location is defined as

$$p_x = (x + 0.5)s, \quad p_y = (y + 0.5)s. \quad (20)$$

The resulting point represents the center of the receptive cell associated with that feature-map location.

For each location, all ground-truth boxes are examined. A box can be assigned to the location only if the location lies inside the box. If this condition is satisfied, the four distances from the location to the box boundaries are computed:

$$l = p_x - x_1, \quad t = p_y - y_1, \quad r = x_2 - p_x, \quad b = y_2 - p_y. \quad (21)$$

The tuple

$$(l, t, r, b) \quad (22)$$

constitutes the regression target for that location.

The second assignment criterion is the regression range associated with the FPN level. In the implemented detector, the ranges are

| FPN level | Regression range |
|-----------|------------------|
| $P_3$     | $[0, 64)$        |
| $P_4$     | $[64, 128)$      |
| $P_5$     | $[128, 256)$     |
| $P_6$     | $[256, 512)$     |
| $P_7$     | $[512, \infty)$  |

(23)

where the size of the object is determined from the maximum of the four regression distances,

$$d_{\max} = \max(l, t, r, b). \quad (24)$$

A location is therefore considered a valid positive sample only if it lies inside the ground-truth box and  $d_{\max}$  falls within the regression range of the current FPN level. This multi-level assignment strategy follows the principle adopted by FCOS, where different feature levels are responsible for different object-size ranges [2]. The corresponding ranges are also explicitly defined in the implementation [2].

When more than one ground-truth box satisfies these conditions for the same location, an ambiguity arises because a single location cannot simultaneously regress multiple targets. The implemented target generator

resolves this ambiguity by selecting the matching ground-truth box with the smallest area. This is consistent with the ambiguity-resolution rule described in FCOS [2].

Once the ground-truth box has been selected, the centerness target is computed from the four LTRB distances as

$$\text{centerness} = \sqrt{\frac{\min(l, r)}{\max(l, r)} \cdot \frac{\min(t, b)}{\max(t, b)}}. \quad (25)$$

The centerness value is maximized when the location lies close to the center of the bounding box and decreases towards its boundaries. Consequently, it provides a measure of how representative a location is for the object being regressed.

For each feature level, the target generator therefore returns

$$\mathcal{T}_i = \{M_i, L_i, C_i\}, \quad (26)$$

where  $M_i$  is the positive-location mask,  $L_i$  contains the LTRB regression targets, and  $C_i$  contains the centerness targets.

## 5. Loss Functions

The detector is trained through three complementary objectives corresponding to the three outputs produced by each detection head: classification, bounding-box regression, and centerness. The classification branch determines whether a spatial location corresponds to a pneumonia instance, the regression branch estimates the spatial extent of the corresponding bounding box, and the centerness branch estimates the quality of the location with respect to the center of the object.

The total training objective is defined as

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}} + \lambda_{\text{ctr}}\mathcal{L}_{\text{ctr}}, \quad (27)$$

where  $\lambda_{\text{cls}}$ ,  $\lambda_{\text{reg}}$ , and  $\lambda_{\text{ctr}}$  weight the contribution of the three objectives. In the present implementation, all three weights are set to one.

The overall formulation is strongly inspired by FCOS [2]. In particular, the classification objective uses the Focal Loss introduced by Lin et al. [5], the bounding-box regression uses Generalized IoU (GIoU) as in the final FCOS formulation, and the centerness target follows the geometric formulation introduced by FCOS. The main modification with respect to the original FCOS objective concerns the loss used for the centerness branch: the present implementation uses the same sigmoid Focal Loss employed for classification instead of binary cross-entropy.

### 5.1. Classification Loss: Sigmoid Focal Loss

The classification branch produces one logit for every spatial location of every FPN level. Because the detector operates densely, only a small fraction of these locations correspond to foreground objects, whereas the majority represent background. This produces a substantial foreground-background imbalance during optimization.

To address this issue, the detector uses the sigmoid Focal Loss introduced by Lin et al. [5] and adopted by FCOS for the classification task [2]. The main intuition behind Focal Loss is to reduce the contribution of examples that are already classified correctly with high confidence and to focus the optimization on difficult or ambiguous examples.

Let  $z$  be the classification logit and let

$$p = \sigma(z) \quad (28)$$

be the corresponding probability obtained using the sigmoid function. For a binary target  $y \in \{0, 1\}$ , the probability assigned to the ground-truth class can be written as

$$p_t = \begin{cases} p, & y = 1, \\ 1 - p, & y = 0. \end{cases} \quad (29)$$

The focal loss is then defined as

$$\mathcal{L}_{\text{focal}} = -\alpha_t(1 - p_t)^\gamma \log(p_t), \quad (30)$$

where  $\alpha_t$  is the class-balancing factor and  $\gamma$  controls the strength of the modulation applied to easy examples.

The implementation uses the standard FCOS/RetinaNet parameters

$$\alpha = 0.25, \quad \gamma = 2. \quad (31)$$

The role of the modulating term can be understood by considering two situations. If a sample is correctly classified with high confidence,  $p_t \approx 1$ , and therefore

$$(1 - p_t)^\gamma \approx 0. \quad (32)$$

Its contribution to the loss becomes small. Conversely, when the prediction is uncertain or incorrect,  $p_t$  is small and the modulation remains large, causing the example to contribute more strongly to the optimization.

In the implementation, binary cross-entropy is first evaluated at every location using logits. The predicted probabilities are then computed with the sigmoid function, followed by the computation of  $p_t$ ,  $\alpha_t$ , and the focal modulation. The resulting per-location losses are summed and normalized globally at a later stage.

## 5.2. Bounding-Box Regression Loss: Generalized IoU

The regression branch predicts four non-negative distances for each positive location,

$$(l, t, r, b), \quad (33)$$

representing the distances from the feature-map location to the left, top, right, and bottom sides of the corresponding ground-truth bounding box. This representation is inherited from the anchor-free formulation of FCOS [2].

The original FCOS formulation uses an IoU-based regression objective, while GIoU is reported in the paper as an improvement to the IoU loss [2]. The motivation for using an overlap-based objective is that the quality of a bounding-box prediction should be directly related to the overlap between the predicted and ground-truth boxes, rather than only to the distance between their parameter vectors.

For two bounding boxes  $A$  and  $B$ , the Intersection over Union is defined as

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}. \quad (34)$$

IoU is intuitive because it directly measures the overlap between the two boxes. However, when the boxes do not overlap,

$$|A \cap B| = 0 \quad \Rightarrow \quad \text{IoU}(A, B) = 0. \quad (35)$$

In this situation, IoU alone does not provide a useful gradient describing how the predicted box should move towards the target. Generalized IoU addresses this limitation [6].

Let  $C$  denote the *smallest enclosing bounding box* containing both  $A$  and  $B$ . In other words,  $C$  is the smallest axis-aligned rectangle that contains the complete predicted box and the complete ground-truth box. Generalized IoU is defined as

$$\text{GIoU}(A, B) = \text{IoU}(A, B) - \frac{|C \setminus (A \cup B)|}{|C|}. \quad (36)$$

The corresponding regression loss is

$$\mathcal{L}_{\text{GIoU}} = 1 - \text{GIoU}(A, B). \quad (37)$$

The first term rewards overlap between the predicted and target boxes. The second term penalizes the portion of the smallest enclosing box  $C$  that is not covered by either box. Therefore, even when the two boxes are disjoint, the loss remains sensitive to their relative spatial configuration. This provides a more informative optimization signal than IoU alone [6].

In the present implementation, the network first produces raw regression outputs, which are passed through a ReLU activation to enforce non-negative distances. The resulting FCOS-style LTRB representation is then decoded into XXYX coordinates before GIoU is computed.

For a location with image-space coordinates  $(p_x, p_y)$ , the decoded box is

$$\hat{x}_1 = p_x - \hat{l}, \quad \hat{y}_1 = p_y - \hat{t}, \quad (38)$$

$$\hat{x}_2 = p_x + \hat{r}, \quad \hat{y}_2 = p_y + \hat{b}. \quad (39)$$

The predicted box can then be compared directly with the corresponding ground-truth box using the GIoU objective.

This aspect of the implementation is therefore particularly close to the final FCOS formulation: both use the FCOS LTRB representation and a Generalized IoU-based regression objective.

### 5.3. Centerness Loss

The centerness branch is used to estimate the quality of a positive spatial location. A location may lie inside a ground-truth bounding box while still being close to one of its boundaries. Such locations generally provide less reliable localization than locations near the center of the object.

FCOS introduces centerness to suppress these lower-quality predictions. The centerness target is computed directly from the four LTRB distances [2]:

$$c^* = \sqrt{\frac{\min(l, r)}{\max(l, r)} \cdot \frac{\min(t, b)}{\max(t, b)}}. \quad (40)$$

If the location is close to the geometric center of the ground-truth box, the left/right and top/bottom distances are similar and the two ratios approach one. Consequently,

$$c^* \approx 1. \quad (41)$$

When the location approaches one of the box boundaries, one of the ratios becomes smaller and the centerness approaches zero. The target therefore acts as a geometric quality measure for the spatial location.

In the original FCOS formulation, the centerness branch is optimized using binary cross-entropy. The present implementation instead uses the sigmoid Focal Loss introduced above:

$$\mathcal{L}_{\text{ctr}} = -\alpha_t(1 - p_t)^\gamma \log(p_t). \quad (42)$$

The use of Focal Loss for centerness represents a deliberate modification of the original FCOS objective. The centerness target remains unchanged and is computed according to the FCOS geometric definition, while the optimization criterion is changed to introduce the same hard-example focusing mechanism used for classification. Unlike the classification branch, centerness loss is evaluated only on positive locations, since background locations do not have a meaningful centerness target.

### 5.4. Global Loss Normalization

The detector predicts over all locations of all FPN levels, but only a fraction of these locations correspond to foreground objects. Furthermore, the number of positive locations can vary from one image or batch to another. Directly summing the losses would therefore make the magnitude of the objective strongly dependent on the number of positive locations.

To reduce this dependence, the implementation counts the total number of positive locations across the complete batch and all FPN levels:

$$N_{\text{fg}} = \sum_b \sum_i N_{\text{fg}}^{(b,i)}. \quad (43)$$

Each raw loss component is then divided by the same normalization factor:

$$\mathcal{L}_k = \frac{\mathcal{L}_k^{\text{raw}}}{\max(1, N_{\text{fg}})}, \quad (44)$$

where

$$k \in \{\text{cls}, \text{reg}, \text{ctr}\}. \quad (45)$$

The final objective can therefore be expressed as

$$\mathcal{L} = \frac{\lambda_{\text{cls}} \mathcal{L}_{\text{cls}}^{\text{raw}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}^{\text{raw}} + \lambda_{\text{ctr}} \mathcal{L}_{\text{ctr}}^{\text{raw}}}{\max(1, N_{\text{fg}})}. \quad (46)$$

The use of a global foreground normalizer keeps the relative scale of the loss components more stable when the number of matched foreground locations changes between batches.

## 5.5. Comparison with the Reference Formulations

The loss formulation adopted in this project is based on the FCOS framework, but differs from both the original FCOS formulation and the reference pneumonia detection study by Wu et al. [1, 2].

| Component                  | FCOS                 | Wu et al.         | This project    |
|----------------------------|----------------------|-------------------|-----------------|
| Classification             | Focal Loss           | –                 | Focal Loss      |
| Regression representation  | LTRB                 | LTRB-based scale  | LTRB            |
| Box regression loss        | IoU Loss             | Smooth L1         | GIoU Loss       |
| Centerness / center target | FCOS centerness      | Center prediction | FCOS centerness |
| Centerness / center loss   | Binary Cross-Entropy | Focal Loss        | Focal Loss      |

## 6. Training Methodology

The training procedure was developed progressively because the ResNet-101 configurations converged more slowly and showed larger validation fluctuations than ResNet-50. The final recipe combines backbone freezing, learning-rate warm-up, cosine annealing, differential learning rates, gradient clipping and exponential moving average (EMA).

### 6.1. Backbone Pretraining and Initialization

For the Chest-Xray initialization, the ResNet backbone was first pretrained on a binary classification task. While the original classification pipeline uses images resized to  $224 \times 224$  pixels, the backbone used in this project was also pretrained at the final detection resolution of  $512 \times 512$  pixels. Training included random horizontal and vertical flips and brightness jitter, while validation and test images were not augmented. The resulting checkpoint was then used only to initialize the convolutional backbone of the detector. For the ImageNet setting, the corresponding pretrained torchvision ResNet weights were used instead.

### 6.2. Backbone Freezing and Learning Rate

At the beginning of detector training, the ResNet backbone is frozen for a configurable number of epochs while the FPN and detection heads are optimized. Once this period is completed, the backbone is unfrozen and trained with a smaller learning rate than the newly initialized detector components. If  $\eta_D$  denotes the FPN/head learning rate, the backbone rate is

$$\eta_B = \alpha_B \eta_D. \quad (47)$$

The standard configuration uses  $\alpha_B = 0.1$ ; R101-IN uses  $\alpha_B = 0.3$ . This difference is intended to preserve useful pretrained features while still allowing the backbone to adapt to the detection task.

The learning rate starts at 1% of the base value and is increased linearly during a warm-up phase. It is then reduced through cosine annealing towards  $\eta_{\min} = 10^{-6}$ :

$$\eta_t = \eta_{\text{base}} \left( 0.01 + 0.99 \frac{t}{T_{\text{warmup}}} \right), \quad (48)$$

followed by

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\text{base}} - \eta_{\min}) \left[ 1 + \cos \left( \frac{\pi t}{T} \right) \right]. \quad (49)$$

The standard warm-up lasts two epochs, while R101-IN uses five.

### 6.3. Gradient Clipping and EMA

To limit unstable optimization steps, the global gradient norm is clipped at 1.0. The training procedure also maintains an EMA of the trainable parameters,

$$\theta_t^{\text{EMA}} = \beta \theta_{t-1}^{\text{EMA}} + (1 - \beta) \theta_t, \quad (50)$$

with  $\beta = 0.999$ . EMA parameters are used during validation and checkpoint selection, while the optimizer continues to update the ordinary model parameters.

## 6.4. Final Training Recipes

The four final configurations use the same detector and evaluation procedure, but the optimization schedule was adapted during development. R50-CX and R50-IN use the initial recipe; R101-CX is trained longer, while R101-IN uses earlier unfreezing, a longer warm-up, a larger batch size and a higher backbone learning-rate factor. Fine-tuning starts from the best checkpoint and uses  $10^{-4}$  for the FPN/heads and  $10^{-5}$  for the backbone. R101-CX was not further fine-tuned because of the limited computational budget and its weaker validation behaviour.

Table 3: Final training recipes.

| Configuration      | R50-CX    | R50-IN    | R101-CX   | R101-IN            |
|--------------------|-----------|-----------|-----------|--------------------|
| Batch size         | 8         | 8         | 8         | 16                 |
| Main training      | 20        | 20        | Extended  | 40                 |
| Frozen backbone    | 3         | 3         | 3         | 2                  |
| FPN/head LR        | $10^{-3}$ | $10^{-3}$ | $10^{-3}$ | $10^{-3}$          |
| Backbone LR        | $10^{-4}$ | $10^{-4}$ | $10^{-4}$ | $3 \times 10^{-4}$ |
| Backbone LR factor | 0.1       | 0.1       | 0.1       | 0.3                |
| Warm-up            | 2         | 2         | 2         | 5                  |
| EMA                | Yes       | Yes       | Yes       | Yes                |
| Gradient clipping  | Yes       | Yes       | Yes       | Yes                |

## 7. Evaluation Metrics

The evaluation is divided into three complementary levels. First, detection quality is measured across confidence thresholds through Precision–Recall analysis, Average Precision and Average Recall. Second, image-level classification behaviour is analysed at a single threshold selected on the validation set. Finally, the spatial quality of the predicted boxes is measured through IoU-based matching and box-level F1-score. During training, validation AP is monitored and the best checkpoint is retained; when EMA is enabled, validation is performed using the EMA parameters.

### 7.1. IoU-Based Matching

Intersection over Union (IoU) measures the overlap between two bounding boxes,

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}. \quad (51)$$

IoU ranges from zero, for disjoint boxes, to one, for perfect overlap. In the present evaluation a prediction is considered a valid spatial match when  $\text{IoU} \geq 0.5$ . Matching is performed independently for each image. Predictions are first sorted by confidence and then matched to their highest-IoU ground-truth box, subject to the threshold and one-to-one matching constraint. A matched prediction is a true positive, an unmatched prediction is a false positive, and an unmatched ground-truth box is a false negative.

This distinction is important because the evaluation is not based only on whether a model assigns a high confidence to an image. A prediction that correctly identifies an abnormal image but places the bounding box in the wrong region does not count as a correct object-level detection under this criterion.

### 7.2. Precision and Recall

Precision and recall summarize the result of the matching procedure. They are defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}. \quad (52)$$

Precision measures how many retained detections are correct, whereas recall measures how many annotated pneumonia regions are recovered. The two metrics therefore emphasize different failure modes: false positives affect precision, while missed annotations affect recall.

Because every prediction has a confidence score, both quantities depend on the threshold used to retain detections. Increasing the threshold usually reduces the number of false positives, but can also remove correct

low-confidence detections. Lowering it has the opposite effect. Evaluating a range of thresholds therefore provides a more complete description than any single operating point.

### 7.3. Precision–Recall Curve and Average Precision

The Precision–Recall curve is obtained by varying the confidence threshold and recording the corresponding pairs (Recall, Precision). Average Precision (AP) condenses this curve into a single scalar. In continuous form it can be interpreted as the area under the curve,

$$AP = \int_0^1 p(r) dr. \quad (53)$$

The implementation instead uses the interpolated 101-point formulation. For  $r \in \{0, 0.01, \dots, 1\}$ ,

$$p_{\text{interp}}(r) = \max_{\tilde{r} \geq r} p(\tilde{r}), \quad (54)$$

and

$$AP = \frac{1}{101} \sum_r p_{\text{interp}}(r). \quad (55)$$

The matching threshold is fixed to  $\text{IoU} = 0.50$ , so the reported quantity is  $AP@0.50$ . Unlike a threshold-dependent metric such as F1, AP summarizes the behaviour of the detector over its complete confidence range. This is why AP is the main metric used for the integrated comparison of the four models.

### 7.4. Average Recall

Average Recall (AR) measures how many ground-truth objects can be recovered when the number of retained detections per image is limited. Predictions are sorted by confidence and only the top  $K$  detections are considered. In this project  $K = 10$ , giving

$$AR@10 = \frac{\text{recalled ground-truth boxes}}{\text{total number of ground-truth boxes}}. \quad (56)$$

AR is useful alongside AP because two models may recover a similar number of targets while assigning different confidence rankings, which can lead to different precision-recall curves.

### 7.5. Object-Size Specific Metrics

The annotated regions vary substantially in size, so performance is also reported separately for medium and large ground-truth boxes. The area thresholds used by the implementation are

$$32^2 \leq A < 96^2 \quad (57)$$

for medium objects and

$$A \geq 96^2 \quad (58)$$

for large objects. This yields  $AP_M$ ,  $AP_L$ ,  $AR_M$  and  $AR_L$ . The categories are defined from the ground-truth area, not from the predicted box, and are evaluated using the same IoU matching procedure as the overall metrics.

These measures are particularly informative for this dataset. The final experiments show that medium regions are substantially more difficult to detect than large ones, suggesting that the spatial resolution available to the detector remains an important limitation even with a multi-level FPN.

### 7.6. Image-Level Operating Point

Detection metrics such as AP evaluate the complete confidence range. For an image-level decision, however, one threshold must be selected. A threshold  $\tau$  is applied to the retained prediction scores and an image is considered positive when at least one prediction survives the filtering step.

The operating point is chosen on the validation set using Youden’s criterion,



$$J = \text{Sensitivity} + \text{Specificity} - 1, \quad \tau^* = \arg \max_{\tau} J(\tau). \quad (59)$$

The selected  $\tau^*$  is then kept fixed for the subsequent image-level analysis. Precision, recall, specificity and F1-score are computed together with the corresponding confusion matrix.

This threshold calibration is conceptually different from AP. AP evaluates the ranking quality of the detector across the whole confidence range, whereas Youden’s  $J$  is used to select one practical operating point that balances sensitivity and specificity.

### 7.7. Localization and Qualitative Analysis

Once the operating point has been selected, the matched boxes provide two additional localization measures: mean IoU over successful matches and box-level F1-score. The average number of retained predictions per image is also reported, since it helps identify whether a model tends to generate multiple candidate regions for the same radiograph.

The quantitative measures are complemented by qualitative overlays of predicted and ground-truth boxes. These examples are useful for distinguishing image-level disagreements from localization disagreements, especially when two models produce the same positive/negative decision but place their boxes differently.

## 8. Experiments and Results

Approximately twenty training runs were performed during development to study backbone depth, initialization, training duration, and optimization settings. The final comparison focuses on four configurations: R50-CX, R50-IN, R101-CX, and R101-IN. All configurations share the same detector architecture, data split, target assignment, and evaluation procedure, while their training schedules were adapted according to the convergence behaviour observed during development.

### 8.1. Training Dynamics

The initial 20-epoch training schedule was sufficient for ResNet-50, whereas ResNet-101 exhibited slower convergence and greater fluctuations in validation performance. The training schedules of the final configurations were therefore adjusted to account for these differences.

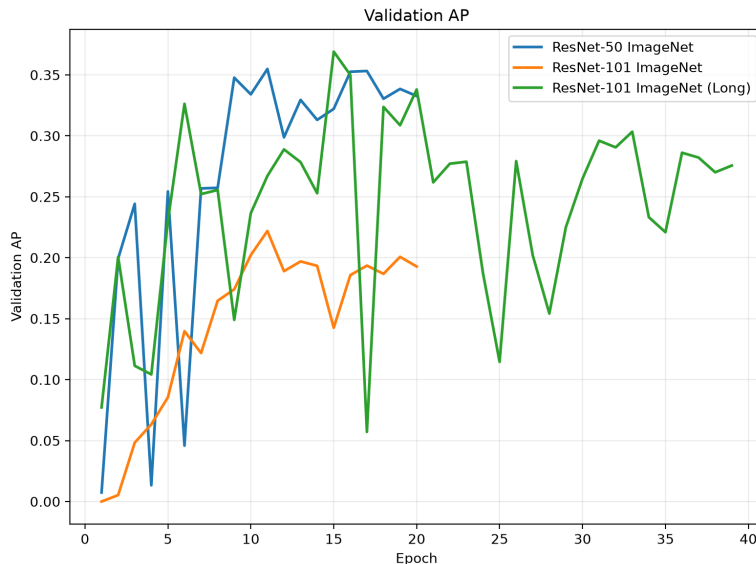


Figure 6: Validation AP during development.

A first extended experiment showed that simply increasing the training time while keeping the original optimization strategy did not provide a large gain. The subsequent R101-IN recipe therefore changed several parameters at the same time: the backbone was unfrozen earlier, warm-up was extended, the batch size was increased from 8 to 16, and the backbone learning-rate factor was raised from 0.1 to 0.3. The model was then fine-tuned starting from the best checkpoint rather than from the final epoch of the main training phase.

This development process is relevant to the interpretation of the final comparison. The four models do not have identical optimization histories, even though they use the same detector architecture and evaluation pipeline. The results should therefore be read as a comparison of the final systems obtained under the available computational budget, rather than as a perfectly controlled ablation of depth and initialization alone.

## 8.2. Final Model Comparison

The final detection results are reported in Table 4. R50-IN achieves the highest AP (0.423), followed by R50-CX (0.399) and R101-IN (0.397), while R101-CX reaches 0.239. Increasing depth alone therefore does not guarantee higher AP. For all configurations, medium objects are considerably harder than large ones; the best  $AP_M$  is 0.039, compared with  $AP_L = 0.442$  for R50-IN.

Table 4: Detection performance of the four final configurations.

| Model   | AP           | $AP_M$       | $AP_L$       | AR@10        | $AR_M$       | $AR_L$       |
|---------|--------------|--------------|--------------|--------------|--------------|--------------|
| R50-CX  | 0.399        | 0.024        | 0.433        | 0.876        | 0.904        | 0.981        |
| R50-IN  | <b>0.423</b> | <b>0.039</b> | <b>0.442</b> | 0.886        | 0.922        | 0.988        |
| R101-CX | 0.239        | 0.029        | 0.231        | 0.804        | 0.920        | 0.974        |
| R101-IN | 0.397        | 0.035        | 0.413        | <b>0.901</b> | <b>0.924</b> | <b>0.989</b> |

R101-IN has the highest  $AR@10$  (0.901), followed by R50-IN (0.886), and also gives the highest recall for medium and large objects. This is a useful complement to the AP ranking: the deeper ImageNet model recovers more ground-truth regions when up to ten detections are allowed, even though its confidence ranking is slightly less favourable overall. R101-CX has the lowest  $AR@10$  (0.804), although its size-specific recall remains relatively high.

The difference between AP and AR is also visible in the size-specific results. All models perform much better on large regions than on medium ones, with  $AP_M$  remaining below 0.04 for every configuration. The limitation is therefore not restricted to a single backbone: small or medium annotated regions are intrinsically harder for the present detector.

The Precision–Recall curves in Figure 7 confirm the AP ranking. R50-IN provides the strongest integrated precision–recall trade-off, while R101-CX remains below the other configurations over most of the recall range.

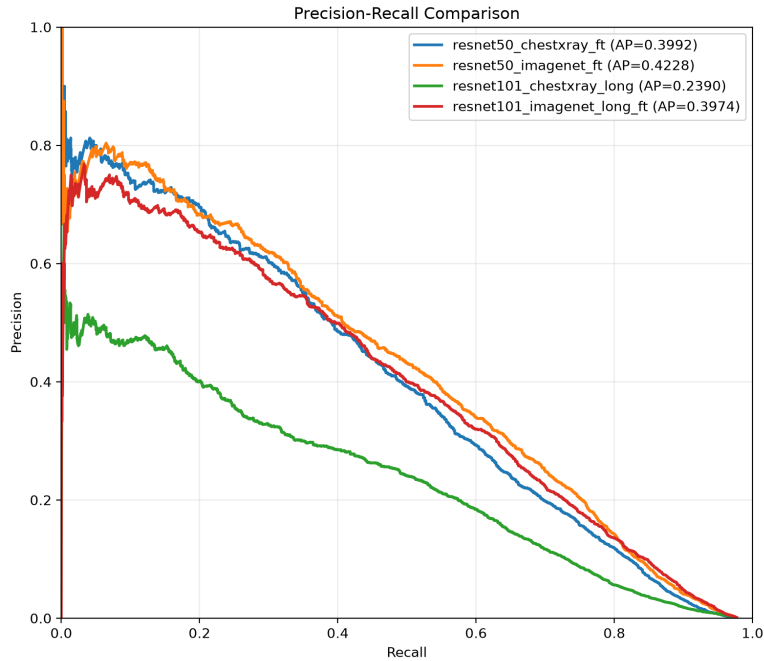


Figure 7: Precision–Recall curves.

**Interpretation of the final comparison.** The final results suggest that model depth mainly improves the ability to recover annotated regions rather than automatically improving the integrated precision–recall score. Moving from R50-IN to R101-IN increases  $AR@10$  from 0.886 to 0.901, while AP decreases slightly from

0.423 to 0.397. The same pair also has very similar box-level F1 scores (0.448 and 0.446). This points to a difference in confidence ranking and detection behaviour rather than a large difference in raw box localization quality.

The Chest-Xray initialization provides a useful contrast. R50-CX remains close to the ImageNet models in several recall measures and reaches  $AP = 0.399$ , whereas R101-CX drops to  $AP = 0.239$ . Since the final training schedules were not identical, especially for the longer R101 experiments, this result is better interpreted as a property of the complete training pipelines than as an isolated measurement of pretraining quality.

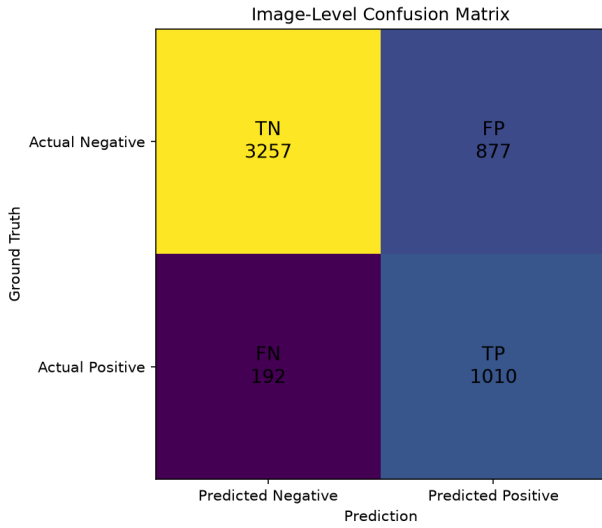
### 8.3. Operating-Point Analysis

The model-specific operating points are reported in Table 5. The selected thresholds are close (0.36–0.41), so the differences are not simply due to very different confidence regions. R101-IN is strongest at this operating point, with precision 0.552, recall 0.846, specificity 0.800, F1 0.668 and Youden  $J = 0.646$ .

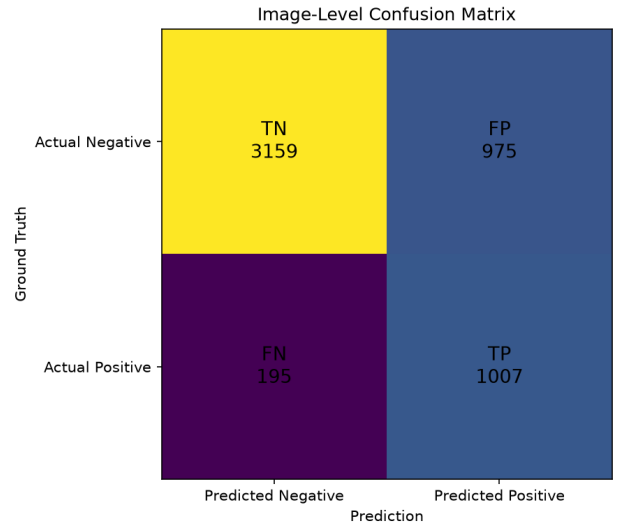
Table 5: Image-level performance at the Youden operating point.

| Model   | $\tau^*$ | Precision    | Recall       | Specificity  | F1           | Youden J     |
|---------|----------|--------------|--------------|--------------|--------------|--------------|
| R50-CX  | 0.36     | 0.508        | 0.838        | 0.764        | 0.633        | 0.602        |
| R50-IN  | 0.39     | 0.535        | 0.840        | 0.788        | 0.654        | 0.628        |
| R101-CX | 0.38     | 0.535        | 0.824        | 0.792        | 0.649        | 0.616        |
| R101-IN | 0.41     | <b>0.552</b> | <b>0.846</b> | <b>0.800</b> | <b>0.668</b> | <b>0.646</b> |

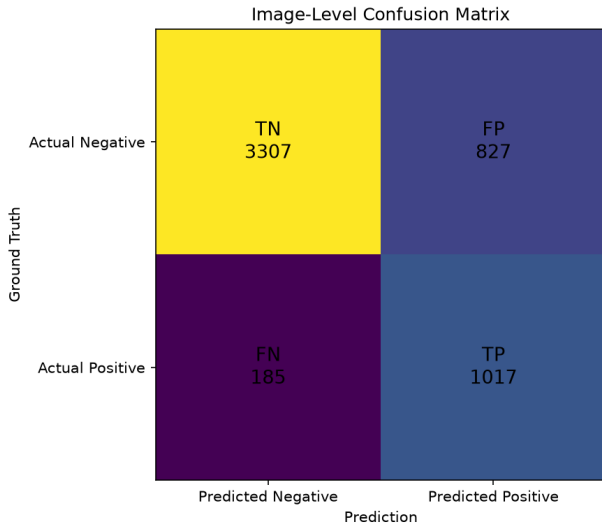
The corresponding confusion matrices are shown in Figure 8. R101-IN produces the largest number of true positives (1017) and true negatives (3307), as well as the fewest false positives (827). These results also illustrate why AP and image-level F1 need not give the same ranking.



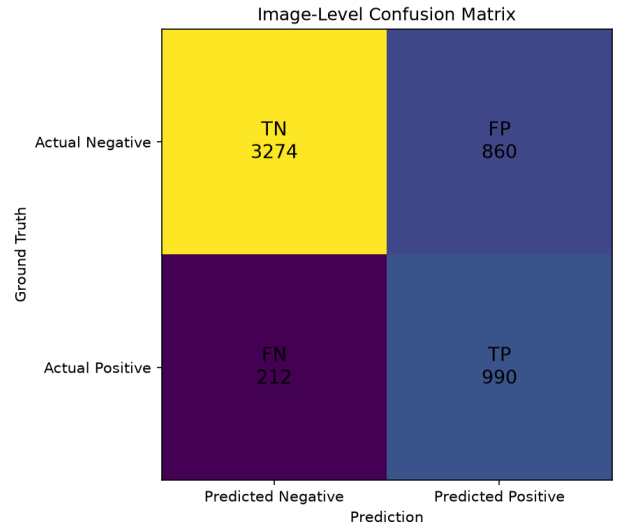
(a) R50-IN



(b) R50-CX



(c) R101-IN



(d) R101-CX

Figure 8: Confusion matrices at the Youden operating points.

#### 8.4. Localization Analysis

The mean matched IoU is 0.684, 0.688, 0.659 and 0.692 for R50-CX, R50-IN, R101-CX and R101-IN, respectively. R101-IN therefore gives the highest average overlap. Box-level F1 is 0.424, 0.448, 0.388 and 0.446, showing that the two ImageNet configurations are close at box level.

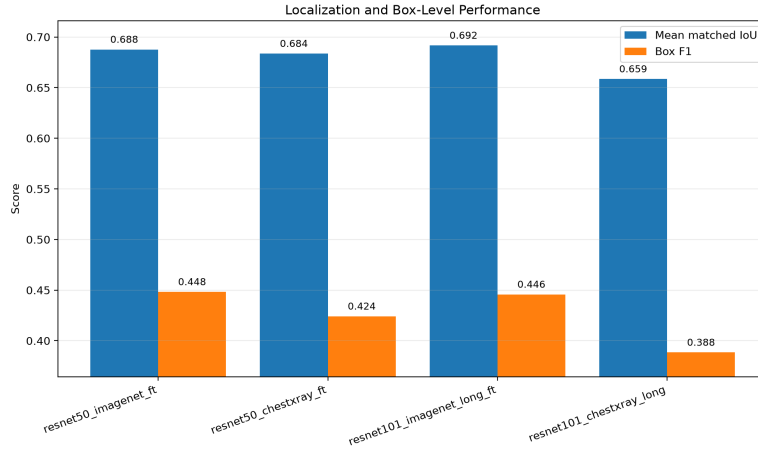


Figure 9: Localization metrics at the Youden operating points.

R101-IN’s lower AP does not correspond to poorer localization: it combines the highest mean matched IoU with the highest image-level recall, while R50-IN remains preferable when integrated AP is the main criterion.

## 8.5. Qualitative Analysis

The qualitative examples are evaluated at the model-specific Youden thresholds and illustrate both image-level and localization disagreement. In Figure 10, (a) R101-CX misses the case while the other configurations detect it, whereas in (b) both ImageNet-initialized models miss the annotated target while the Chest-XRay-initialized models detect it. The localization examples in Figure 11 show that, even when the image-level decision agrees, the predicted boxes can differ in number, size, and position.

These examples highlight two complementary aspects of detector behaviour. Image-level disagreement reflects whether a prediction is retained at all, whereas localization disagreement concerns differences in the spatial extent and position of the predicted boxes. This supports the use of both detection metrics and box-level analysis when comparing the four configurations.

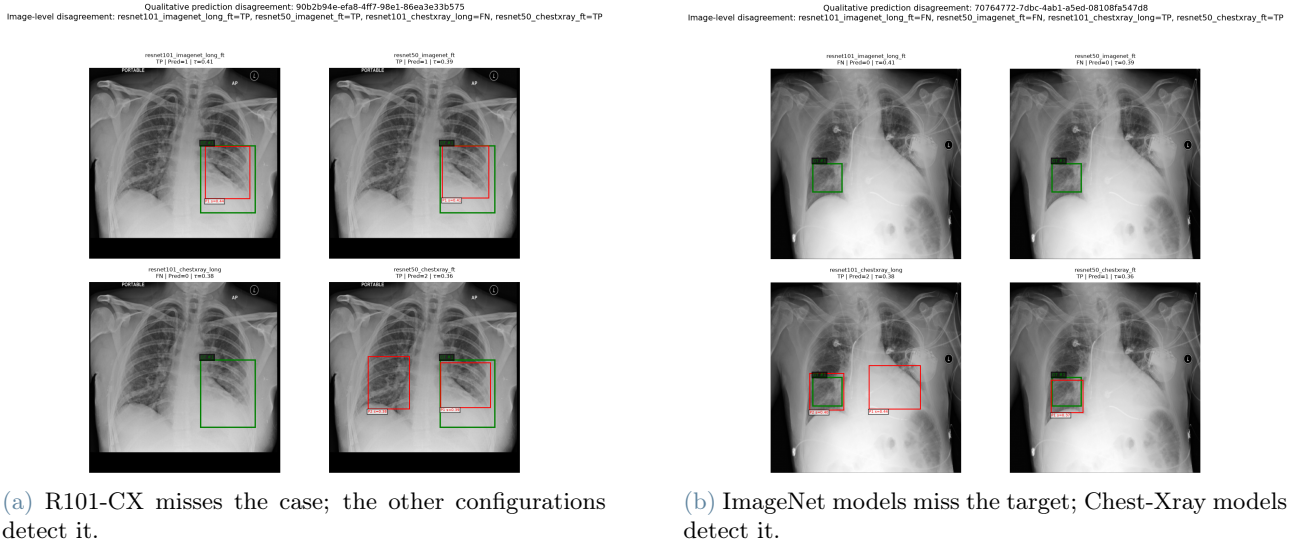
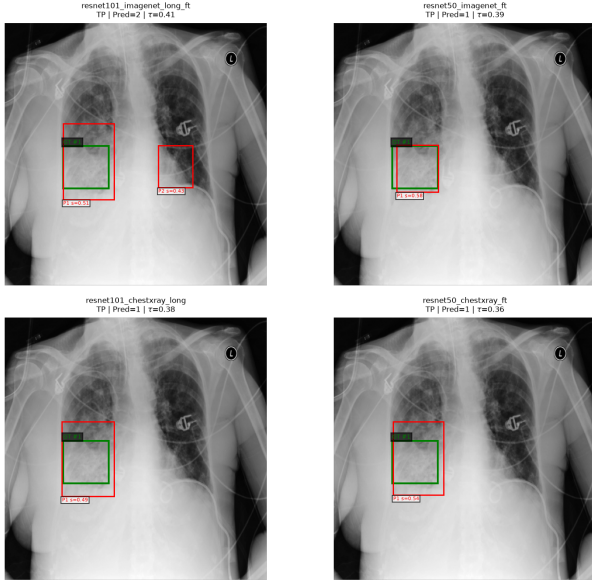


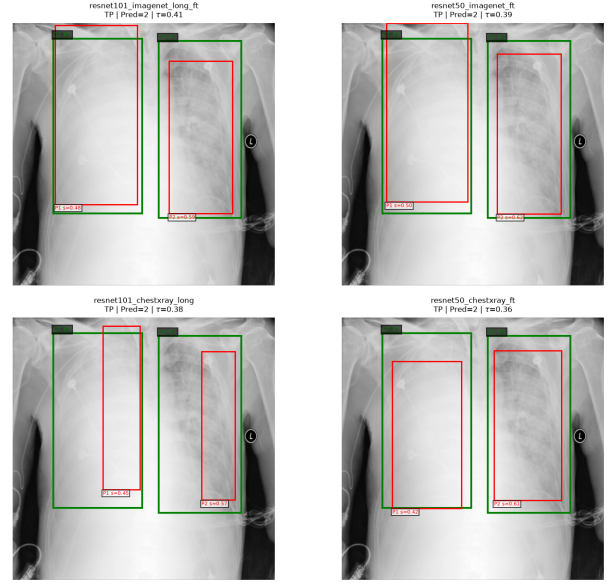
Figure 10: Representative image-level prediction disagreements. Green: ground truth; red: predictions.

Qualitative prediction disagreement: 5feccc14-e51d-4c12-bc60-4cdf83014ee2  
Pure localization disagreement: all models=TP; CSV IoU gap=0.817; CSV count gap=1

Qualitative prediction disagreement: a20a5357-fcdc-4d8f-b5ad-7a49c2df7db5  
Pure localization disagreement: all models=TP; CSV IoU gap=0.821; CSV count gap=0



(a) Localization disagreement case 1



(b) Localization disagreement case 2

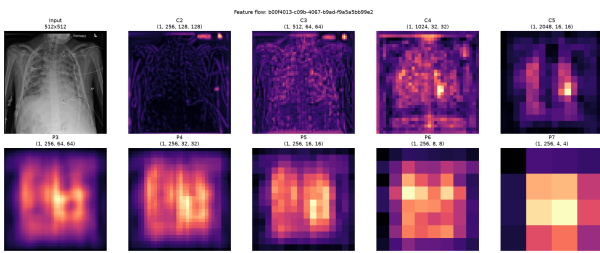
Figure 11: Representative localization disagreements. Green: ground truth; red: predictions.

## 8.6. Additional Visualizations

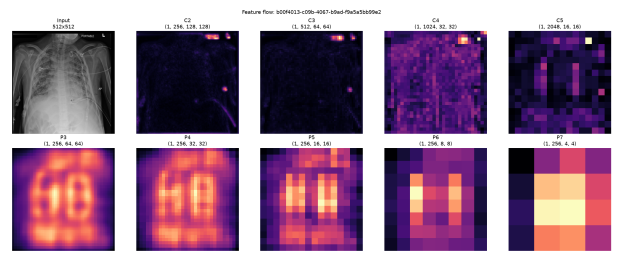
Additional qualitative visualizations are provided to complement the quantitative results and illustrate both the intermediate feature representations and the final detector predictions.

### 8.6.1. Feature Flow

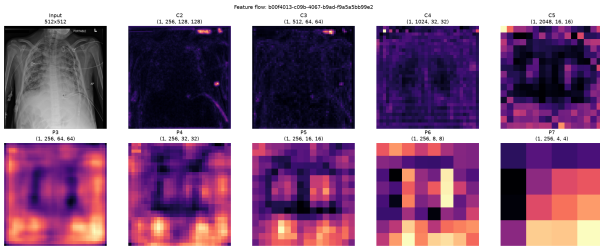
The same input radiograph is passed through the four final configurations, and the resulting feature flow through the backbone and FPN is visualized in Figure 12. These representations are intended for qualitative comparison and do not provide a direct performance measure.



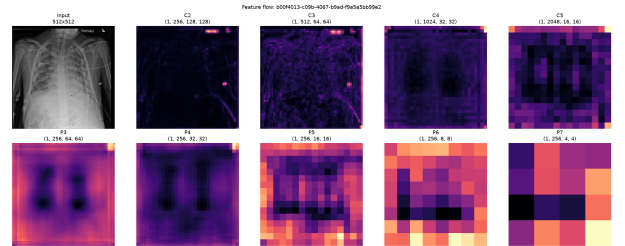
(a) R50-CX



(b) R50-IN



(c) R101-CX

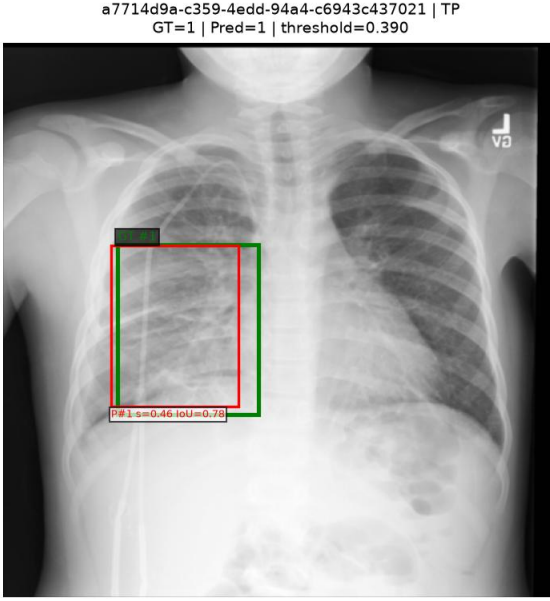


(d) R101-IN

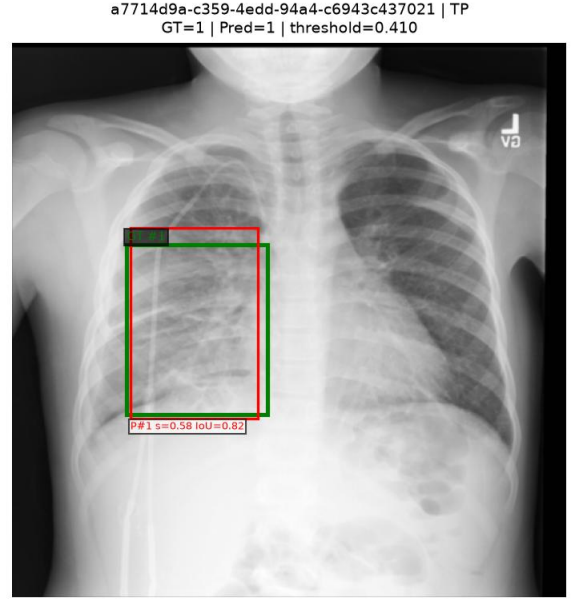
Figure 12: Feature flow for the four final detector configurations on the same input image.

### 8.6.2. Representative Detections

Figures 13 and 14 show representative predictions from the two ImageNet-initialized configurations, R50-IN and R101-IN, selected as the strongest overall models in the quantitative comparison.

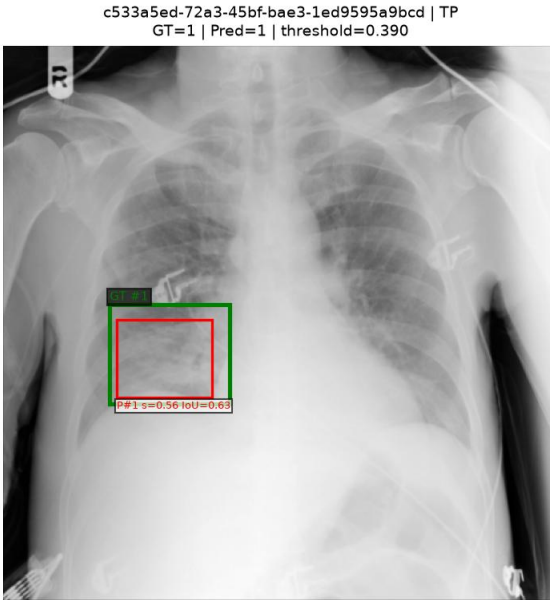


(a) R50-IN – case 1

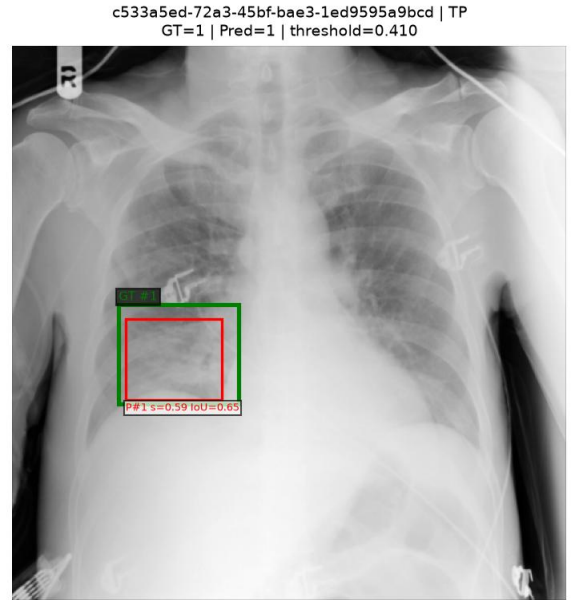


(b) R101-IN – case 1

Figure 13: Representative detections for case 1 produced by R50-IN and R101-IN.



(a) R50-IN – case 2



(b) R101-IN – case 2

Figure 14: Representative detections for case 2 produced by R50-IN and R101-IN.

### 8.7. Conclusions

The results show that no single configuration is best under every criterion. R50-IN achieves the highest AP (0.423), whereas R101-IN provides the highest  $AR@10$  (0.901), the best image-level F1 at the Youden operating point (0.668), and the highest mean matched IoU (0.692). The deeper backbone is therefore not uniformly



better, but it is competitive when recall and localization are emphasized.

The Chest-Xray-initialized models are weaker in the present experiments, especially R101-CX ( $AP = 0.239$ ). This result should not be interpreted as evidence that domain-specific pretraining is intrinsically less effective. The optimization budgets were not fully matched, and the R101 configurations required different training schedules and different amounts of fine-tuning. A controlled comparison would require comparable training budgets, repeated runs, and uncertainty estimates.

## 8.8. Clinical and Dataset Limitations

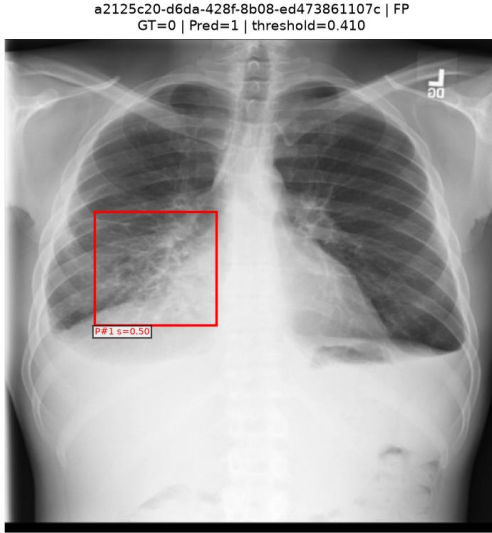
The quantitative results should also be interpreted in the context of the information available in the dataset. A discussion with a radiologist during the qualitative review highlighted several limitations that are difficult to capture using detection metrics alone. In particular, the detector operates on a single chest radiograph, whereas the clinical assessment of suspected pneumonia can make use of additional projections and, when necessary, follow-up examinations. Frontal and lateral views can provide complementary information about the location and extent of an abnormality, while repeated imaging can help distinguish persistent findings from transient ones. None of this additional information is available to the present model.

Patient positioning is another relevant factor. Many examinations in the RSNA dataset are portable radiographs, including images acquired from patients who are not standing. The radiologist consulted for this study emphasized that positioning can alter the apparent distribution and extent of pulmonary opacities. In particular, small opacities may be less evident in supine examinations, while more extensive areas of increased density may appear more diffuse. The visual appearance of a lesion is therefore influenced not only by the underlying pathology but also by acquisition conditions and patient position.

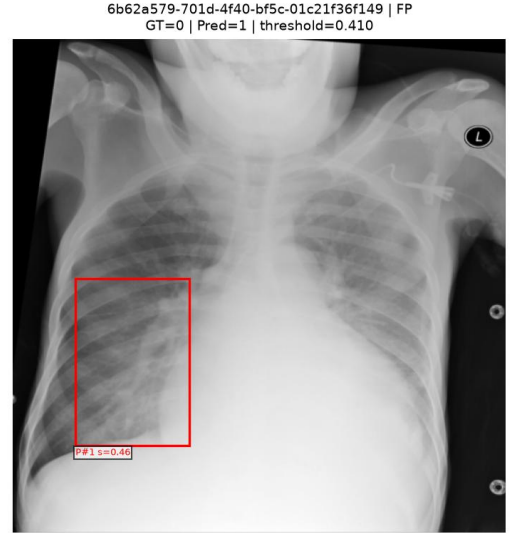
More importantly, the radiographic pattern used by the detector is only one part of the clinical assessment. An opacity compatible with pneumonia can also be produced by other thoracic conditions. The consulting radiologist noted that some neoplastic findings can have an appearance similar to infection on a single chest radiograph. In a clinically uncertain case, the radiographic finding may be described first as an opacity or consolidation, and its meaning is then assessed together with symptoms, medical history and previous imaging. If the abnormality persists despite an initial clinical treatment, additional imaging such as CT may be required to clarify the diagnosis. This means that a prediction that disagrees with a dataset label should not automatically be interpreted as a clinically incorrect observation.

The same issue became evident in the inspection of individual predictions from R101-IN. Figure 15 shows two representative cases selected during the radiological review. In the first example, the model marks an opacity adjacent to an enlarged cardiac silhouette. The opacity itself can look compatible with pneumonia, but the radiologist considered the cardiac silhouette and the overall thoracic appearance essential for interpretation. In this case, the increased density was considered more consistent with a cardiocirculatory condition than with pneumonia. The example illustrates that a model focusing mainly on the local appearance of an opacity may miss information contained in the surrounding anatomy, such as the heart size.

In the second example, R101-IN predicts pneumonia, while the consulting radiologist considered the overall appearance more compatible with atelectasis. The distinction was not based on the opacity alone. Other radiographic signs, including changes involving the costophrenic angles and the diaphragm, were considered together with the appearance and extent of the opacity. These signs are not explicitly represented in the current prediction target, which is primarily based on the location and size of the annotated bounding box.



(a) Opacity adjacent to an enlarged cardiac silhouette.



(b) R101-IN prediction considered compatible with atelectasis.

**Figure 15:** Two R101-IN predictions discussed during the radiological review. The examples illustrate cases in which interpretation depends on the global radiographic appearance and on findings beyond the annotated opacity itself.

These cases also highlight a limitation of using the confidence associated with a single predicted box as a direct proxy for diagnostic certainty. A large or high-confidence prediction is not necessarily more clinically plausible than a smaller one if the surrounding radiographic findings point towards another explanation. Conversely, an extensive pneumonia is possible, but its clinical plausibility still depends on the complete presentation. Future systems could therefore benefit from calibrating the interpretation of detection confidence with the spatial extent of the predicted region and with additional anatomical features rather than relying on the box score alone. The discussion also raises a question about the quality of the ground truth. The RSNA dataset is strongly imbalanced at image level, but the appropriate response is not necessarily to give greater weight to all positive examples. If a subset of positive or negative cases is clinically ambiguous, simple reweighting can increase the influence of uncertain annotations without improving the underlying supervision. A more useful strategy would be to identify and review difficult cases, characterize potentially ambiguous negatives, and, where possible, obtain stronger expert consensus annotations. This would make it possible to separate limitations of the detector from limitations of the reference labels more reliably.

At the same time, the radiological review was positive about the overall quality of the proposed detector. In particular, the radiologist considered false positives to be less concerning, on average, than missed findings in the context of pneumonia screening. This observation does not imply that false positives are clinically irrelevant, but it supports the importance of maintaining high sensitivity when choosing an operating point. The present results are consistent with this perspective: R101-IN does not achieve the highest AP, but it provides the highest recall and the strongest image-level balance at the selected operating point.

Overall, the project shows that the FCOS-style detector is able to provide a compact anchor-free framework for pneumonia localization and that backbone depth, initialization and optimization have a measurable effect on the results. The clinical review, however, shows that localization from a single radiograph is only a partial approximation of the diagnostic process. A more complete system could combine frontal and lateral radiographs when available, anatomical measurements such as cardiac and diaphragmatic information, selected clinical history, and additional examinations such as CT in difficult cases. Such a system would use the detector as a spatially grounded component rather than treating its output as a complete diagnosis.

The main lesson is therefore not simply that a deeper or differently initialized backbone is preferable. The results suggest that improvements in the data and in the information presented to the model may be at least as important as increasing model complexity. Future experiments should first make the training comparison more controlled and the clinical characterization of the dataset more rigorous. This would provide a stronger basis for determining which architectural improvements genuinely improve pneumonia localization and which apparent errors are instead related to the ambiguity of the underlying radiographic evidence.

## 9. Code and Dataset Availability

The complete implementation of the proposed detection pipeline, including the backbone, FPN, detection heads, training procedures, and evaluation scripts, is available in the project repository:

<https://github.com/Francesco2035/fcos-pneumonia-detection-naml-25-26>

The experiments were conducted using two publicly available chest X-ray datasets. The main detection experiments use the RSNA Pneumonia Detection Challenge dataset, which provides chest radiographs together with pneumonia annotations and bounding boxes. The dataset is available through the official RSNA challenge resources.

For backbone pretraining, the Chest-Xray initialization was obtained from the public “Chest X-Ray Images (Pneumonia)” dataset, containing Normal and Pneumonia images for binary classification.

The corresponding dataset resources are available at:

- RSNA Pneumonia Detection Challenge: [RSNA official dataset page](#)
- Chest X-Ray Images (Pneumonia): [Kaggle dataset page](#)

## 10. Acknowledgements

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